

AHMED KHAN PATAN

Dresden, Germany — M.Sc. Computational Modeling and Simulation, TU Dresden



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PROFESSIONAL SUMMARY

Python software engineer and TU Dresden M.Sc. student with approximately 15 months of professional experience in backend development, data processing, APIs, PostgreSQL, automation, cloud systems, and production debugging. Experienced in building structured data workflows and AI-assisted engineering processes, with academic ML projects and strong interest in AI/data engineering and applied research.

PROFESSIONAL EXPERIENCE

ProjectXpert (Project Expert Ventures India Pvt. Ltd.)

Jul 2025 – Aug 2026

Associate Developer

Remote

- Built Python-based automation and data-processing workflows spanning structured files, REST APIs, PostgreSQL-backed applications, and AWS services.
- Developed and maintained Django/Flask services with validation, structured persistence, API integrations, and production troubleshooting.
- Designed an **8-system automation pipeline across 2 AWS accounts** and validated data movement using logs, database checks, and end-to-end test scenarios.
- Worked with Git/GitHub, Docker, Linux, GitHub Actions CI/CD, PostgreSQL, Redis, JSON/XML processing, and AI-assisted development using Claude and GitHub Copilot.

ProjectXpert (Project Expert Ventures India Pvt. Ltd.)

May 2025 – Jun 2025

Software Development Intern

Remote

- Built Python scripts to clean, transform, validate, and process Excel workbooks into structured operational outputs.

SELECTED ACADEMIC PROJECTS

IoT-Enabled Arrhythmia Classification using Deep Learning

Python, CNN, ESP32, IoT

- Developed an IoT-enabled arrhythmia-classification prototype combining ESP32-based physiological sensing, data preprocessing, and CNN-based classification.
- Integrated sensing and software analysis into an end-to-end academic workflow for near-real-time monitoring and prediction.

Phishing Website Detection using Machine Learning

Python, ML, Classification

- Developed a machine-learning workflow to classify websites as legitimate or potentially phishing using extracted website and URL characteristics.
- Prepared features, trained a classification model, and evaluated prediction behavior on labeled data.

EDUCATION

Technische Universität Dresden

Oct 2026 – Sep 2028 (Expected)

M.Sc. Computational Modeling and Simulation

Dresden, Germany

Muffakham Jah College of Engineering and Technology

Sep 2021 – Jul 2025

B.E. Information Technology – CGPA: 8.13/10

Hyderabad, India

TECHNICAL SKILLS

Programming/Data: Python, SQL, PostgreSQL, JSON/XML, Excel data processing

Backend: Django REST Framework, Flask, REST APIs, data validation and integration

Engineering: Git, GitHub, Docker, Linux, GitHub Actions, CI/CD

AI/ML: CNN classification, ML coursework, AI-assisted development with Claude/GitHub Copilot

Cloud: AWS (Lambda, ECS, RDS, S3, CloudWatch)

Languages: English – Professional working proficiency; German – Beginner, currently learning

HOBBIES & INTERESTS

Hobbies: Badminton, Carrom, Travel — **Technical interests:** Cloud systems, AI/ML experimentation, developer tooling